

Inference-Time Diversity in RL-Trained Lean Theorem Provers

A Diagnostic Study

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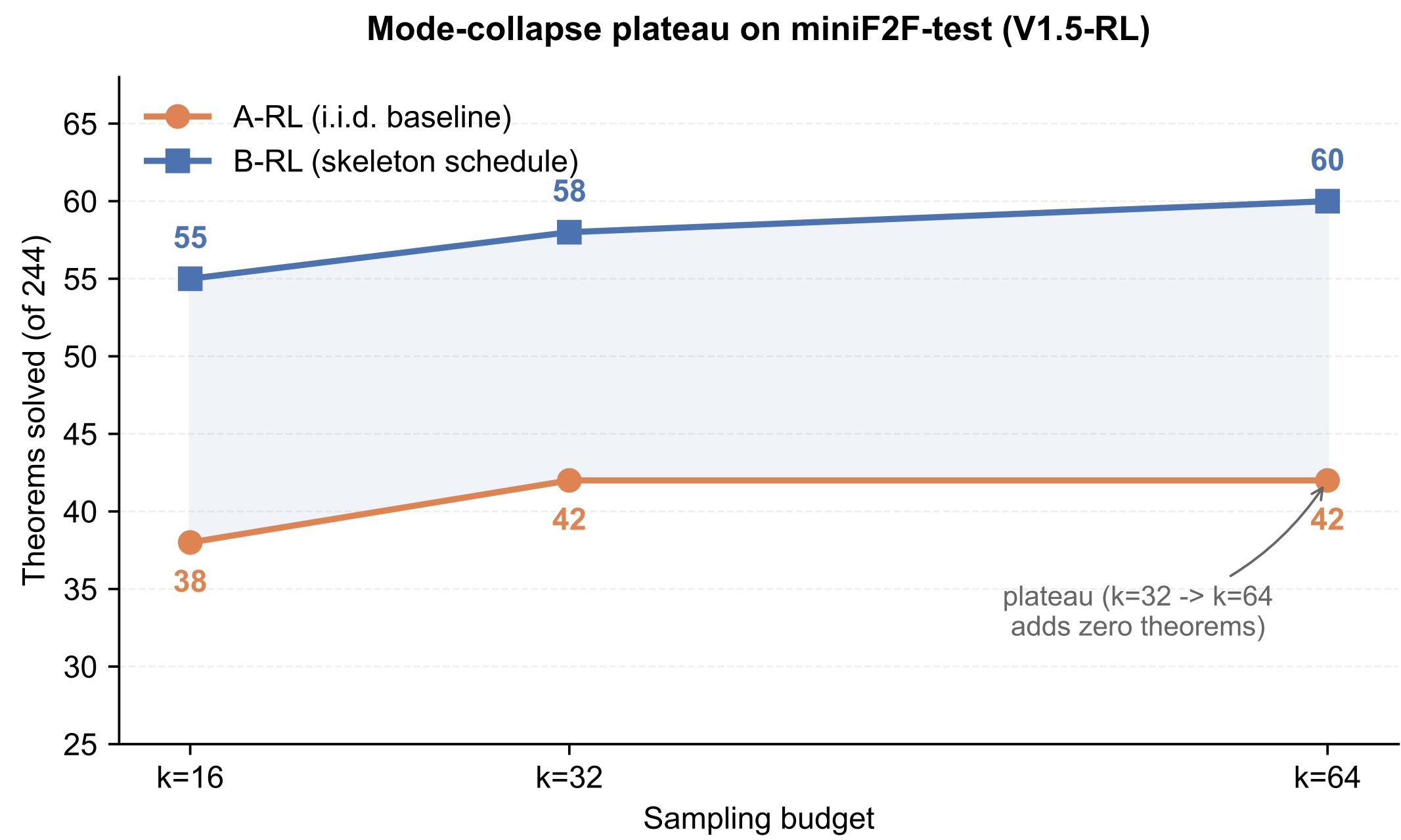


TL;DR *RL-trained Lean theorem provers run out of ideas as you sample more. A fixed schedule of 15 tactic skeletons closes the gap, and the effect is RL-specific.*

Setup

Model: DeepSeek-Prover-V1.5-RL (7B, paired with non-RL V1.5-Base).
Benchmark: miniF2F-test (244 Lean 4 / Mathlib theorems).
Schedule: 15 hand-picked tactic skeletons (simp, intro, constructor, aesop, ...) $\times$ 8 NL goal hints = 120-slot grid. Budgets $k \in \{16, 32, 64\}$ take the first k slots.
Decoding: temperature 0.6, top- p 0.95, max 1024 tokens. **Verifier:** lake env lean -json, 120s timeout, 8-way sharding by $i \bmod 8$.

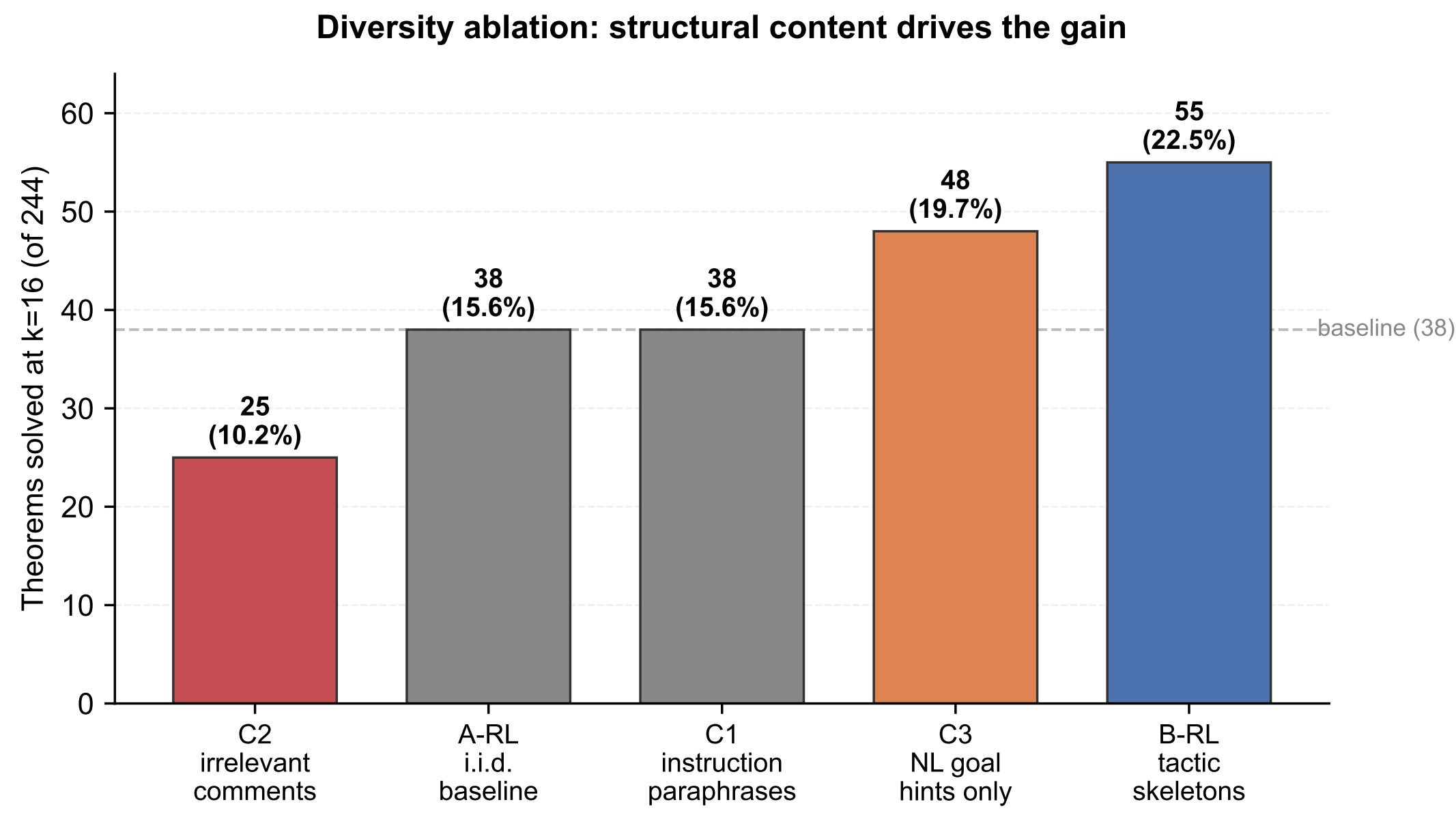
Mode collapse: A-RL plateaus at $k=32$



	$k=16$	$k=32$	$k=64$
A-RL (i.i.d.)	38/244 (15.6%)	42/244 (17.2%)	42/244 (17.2%)
B-RL (skeleton)	55 (22.5%)	58 (23.8%)	60 (24.6%)
Δ absolute	+17	+16	+18
Δ relative	+44.7%	+38.1%	+42.9%

Across-seed: mean $\Delta = +12.3 \pm 4.2$ theorems at $k=16$ ($n=3$ seeds, sign positive every seed).

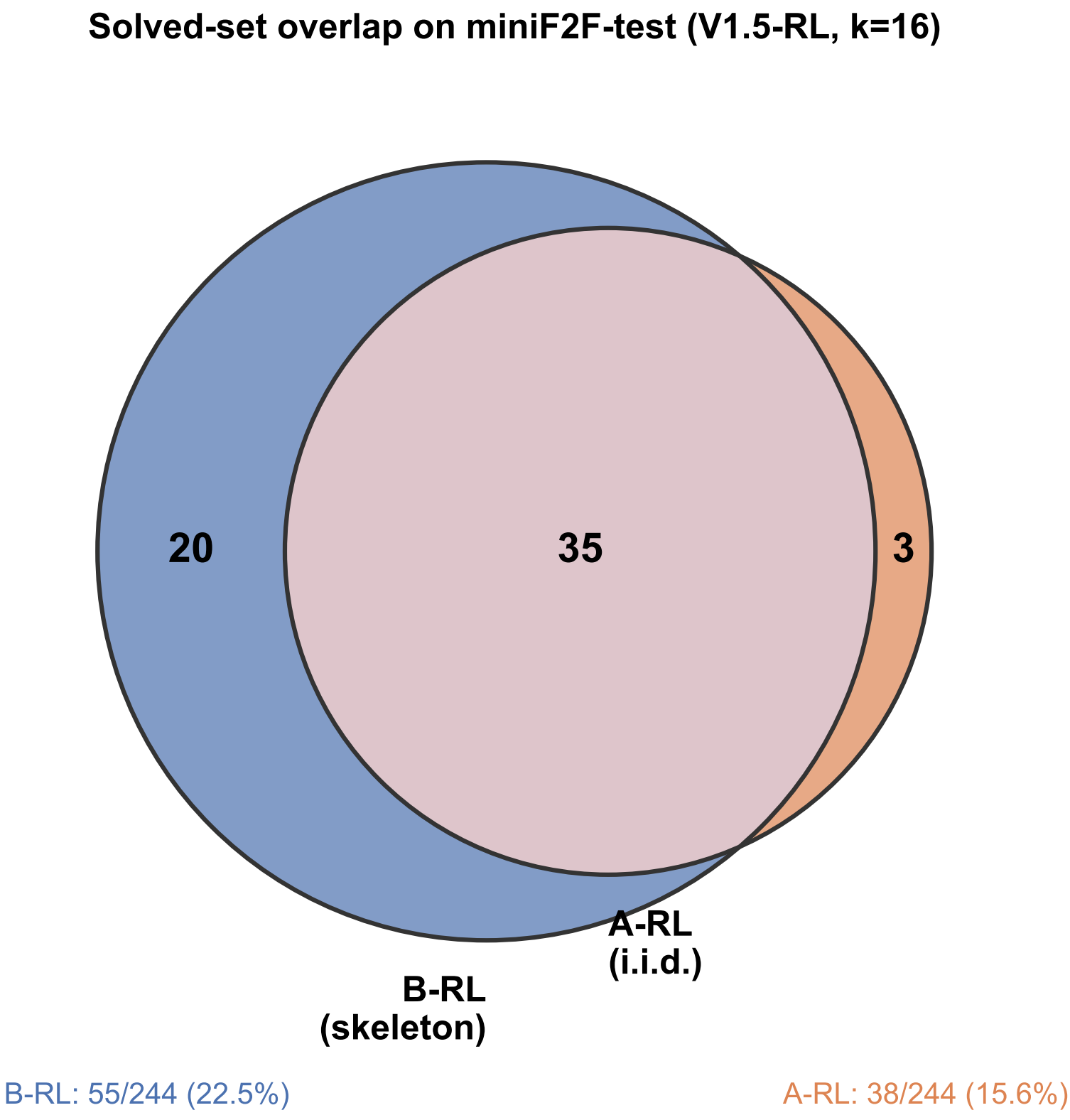
Diversity ablation: structural content drives the gain



C2 < A-RL = C1 < C3 < B-RL. Irrelevant Lean comments *hurt*; paraphrased instructions match the baseline; NL goal hints recover 58% of the $A \rightarrow B$ gap; tactic skeletons go furthest. The gain is structural, not just diversity.

Distributional evidence of collapse: Across **13,159** stochastic V1.5-RL samples, the median theorem gets only **2** distinct first-tactic heads; **49%** of the benchmark gets a deterministic single strategic opening.

Solved-set overlap (V1.5-RL, $k=16$)



B-RL nearly Pareto-dominates A-RL: **20** theorems are solved only by B-RL, **3** only by A-RL, **35** by both.

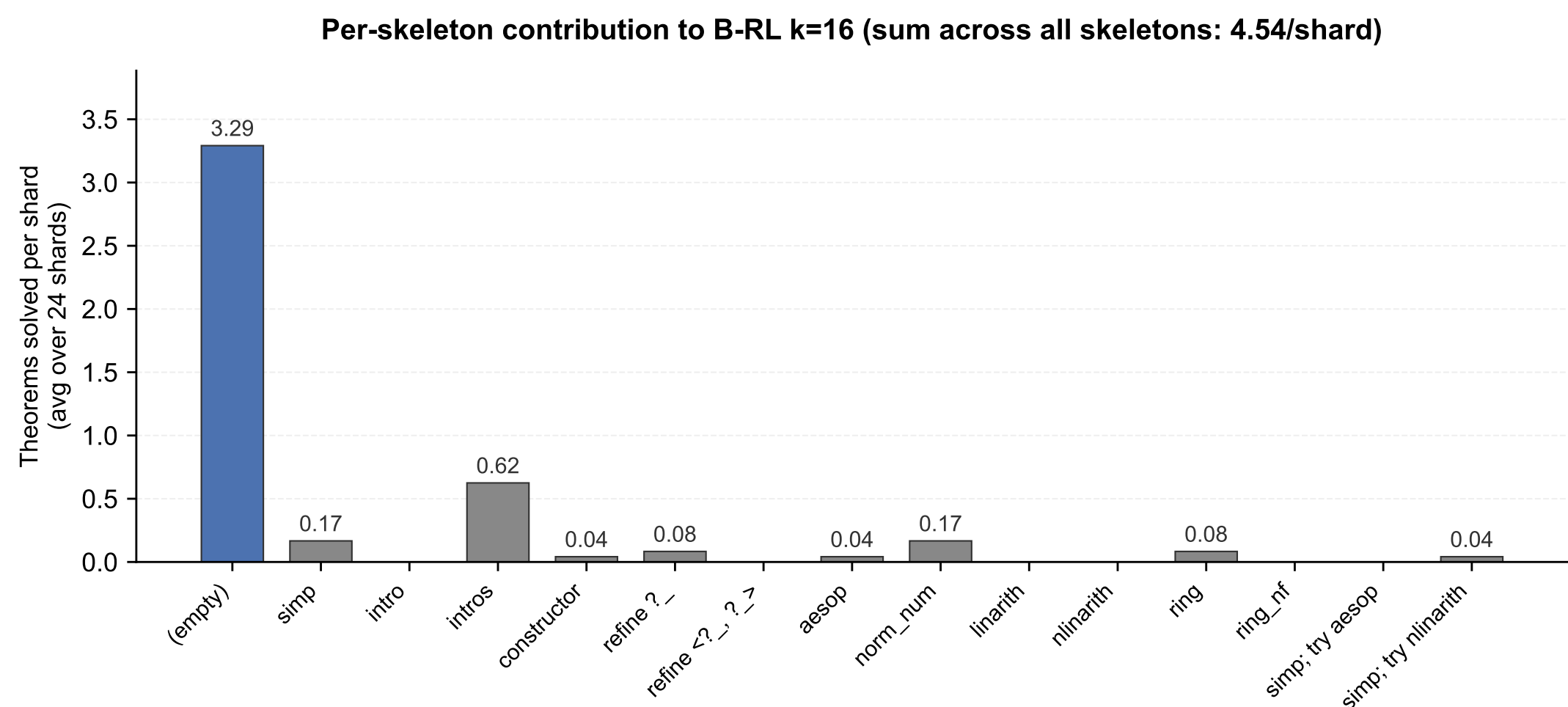
Phenomenon is RL-specific

DeepSeek-Prover-V1.5-Base (the same model, no RL fine-tune) proves **0/244** theorems at every k , with *or* without skeletons.

- Base outputs sorry or echoes the statement in 58.8% of attempts.
- RL model emits sorry in $< 0.2\%$ of attempts; its mode collapse manifests as persistent generation of failing-but-valid rw/apply sequences.

RL is the locus of both the proof capability and the inference-time collapse that limits it.

Per-skeleton contribution (within $k=16$)



24 B-RL $k=16$ shards, within-shard counting. Empty skeleton carries most of the load (3.29/4.54 solves per shard); the other 14 skeletons together pick up the long tail — exactly the 17 theorems A-RL i.i.d. misses.

Cross-model generalisation

$A \rightarrow B$ contrast on two further 7B Lean provers:

- RL-trained V2-7B:** +3 *frontier* theorems no i.i.d. baseline of any tested model reaches.
- SFT Goedel-Prover:** -10.0 ± 4.4 theorems ($n=3$ seeds, negative every seed). The intervention helps RL and *hurts* SFT.

